

dark line represents how the portfolio starts out earning an APR of 5% on the first day of retirement. To be exact, this is 2 basis points (or 0.02%) during the first day of retirement, which is a 5% APR divided by 250 trading days. The market then moves into a bull-market cycle so that your annualized returns slowly increase until it peaks at 20% per annum (8 basis point per day) in approximately 19 months. In the language of sine waves, the market peaks after approximately $\pi/2$ years. Remember the Greek letter π is equal to approximately 3.14 years, which is 37.68 months, so $\pi/2$ is just shy of 19 months. Throughout this scenario, \$7 per year is also being withdrawn.

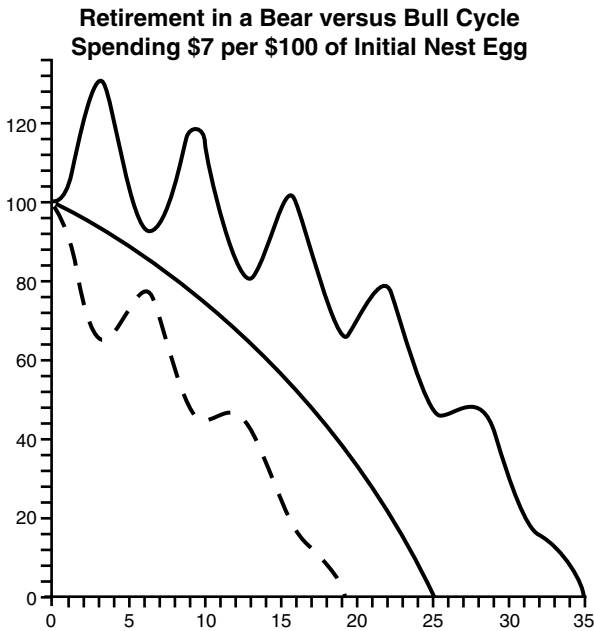


Figure 6.2 Will your runny nest egg zig before it zags?

Source: M. Milevsky, "The Trigonometry of Retirement Income," *Research Magazine*, February 2007.

Then after hitting this peak, financial markets start to decline so that approximately 19 additional months later the market is back to earning an APR of 5%. It has gone from 5% up to 20% and then down to 5% over approximately 3.14 years.